Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	3AM
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 - Quality Education
Project / Solution Name	Pathwaay
Name (Leader)	Ooi Wei Lun
Email & Phone (Leader)	lunweii1025@gmail.com & 0197107401
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is changing the skills needed for employment faster than many students and early-career workers can plan for. The World Economic Forum reports that approximately 39% of workers' existing skill sets may be transformed or outdated by 2030, while 63% of surveyed employers identify skill gaps as a major barrier to business transformation (WEF, 2025). Young adults, jobseekers and people considering career changes may struggle to interpret fragmented job-market information, recognise transferable skills and choose credible training. Generic recommendations also risk obscuring the evidence behind a suggested occupation or course.

For Malaysian learners, these challenges affect access to relevant education and longer-term employability. Pathwaay addresses the planning and information gap which is not unemployment directly by connecting user-reviewed skill evidence with documented occupation requirements, research and vacancy signals, and practical learning options. Its European ESCO reference is explicitly a skills classification, not proof of Malaysian hiring demand. The intended impact aligns primarily with UN SDG 4.4: improving access to skills relevant for employment and decent work.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Pathwaay is a web-based AI-assisted career exploration and reskilling prototype. Users create a profile or review a CV, inspect suggested skills, explore future-role matches and career details, and see skill gaps mapped to documented ESCO occupation requirements. The Role Matching workflow uses a full pinned ESCO v1.2.1 index and transparent weighted skill overlap, while keeping strict essential-skill coverage separate from relevance. Career Details offers explanations with a Personalised Reskilling Pathway links verified learning materials and indicates gaps where resource evidence is insufficient. The AI Career Transition Simulator lets users select exploratory occupation paths and save a personal timeline of sourced skills, goals and breaks.

Other prototype modules offer AI trend signals from research/job-source adapters, a career/SDG scenario demonstration, CV tools and an employer-oriented skills view. The career-model and scenario demonstration use labelled synthetic data, they are not validated job forecasts. Optional RunPod-hosted language-model calls support assisted explanations and skill extraction with user review, Tavily Search/Extract can help discover provider pages where configured, subject to verification.

The solution applies skills from the GenAI, database and cloud workshops through API integration, FastAPI services, SQLite persistence, deterministic source checks and a Next.js interface. Replit and RunPod are represented in the repository, but live deployment and provider availability must be confirmed separately before Phase 2.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16, React 19 and TypeScript, responsive profile, role matching, trend, CV, simulator and employer pages.
Backend: Python, FastAPI, Pydantic, local scikit-learn career model for a labelled synthetic demonstration.
Storage: SQLite for saved fictional profiles, jobs and simulator journeys, validated local ESCO v1.2.1, reviewed resources and source registries.

External integrations: European Commission ESCO, OpenAlex research metadata, Remote OK/Himalayas employment-source adapters, optional Tavily Search/Extract, optional RunPod Serverless LLM endpoints for assisted text/skill interpretation. Replit-compatible development launcher and Next.js/FastAPI API proxy, production hosting and persistence require deployment verification.

SDC sponsor/workshop technology: RunPod Serverless integration is included for optional GenAI processing, the repository contains a RunPod proof-of-use artifact. Replit development support is also configured.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Design thinking followed an iterative problem-to-prototype approach which it to identify uncertainty faced by early-career learners, prototype profile, role-matching, skill-gap and learning screens, test evidence labels, mobile layouts and error recovery, then refine. Users review AI-suggested CV skills before accepting them. ESCO relevance is labelled as skill overlap, not hiring probability or job readiness.

The existing Next.js/FastAPI/SQLite architecture, verified ESCO index, automated tests and browser checks support a functioning prototype. The Career Simulator was tested for save/reload and edit conflicts. Before Phase 2, the team will verify public cloud deployment, durable data storage and provider access, review third-party rights, test additional devices and accessibility, and conduct a pre/post user study of skill-gap identification. Real-person profiles require production authentication and privacy controls.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary UN SDG 4, Target 4.4 which is to increase youth and adults with relevant employment, decent-work and entrepreneurship skills. Supporting SDG 8, Target 8.5 which is decent work for all. Source: https://sdgs.un.org/2030agenda
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	External problem baseline: WEF Future of Jobs Report 2025 estimates 39% of existing skill sets will change by 2030, 63% of employers identify skills gaps as a major barrier. Source: https://www.weforum.org/publications/the-future-of-jobs-report-2025/
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Pilot KP where the percentage of participants who can correctly identify at least 3 ESCO-documented essential skills not yet evidenced for a selected role, with source references, in the same task before versus after Pathwaay. Report baseline, post-use percentage and relative change, also audit source-link accuracy.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 target is at least 20% relative improvement in the KPI after use compared with the pilot pretest baseline. Evaluate with paired pilot tasks and report sample size, failures and uncertainty, avoid claiming an employment-rate impact.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

-

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React/React DOM, TypeScript	Code frameworks / language tools	MIT, MIT, Apache-2.0
FastAPI, Uvicorn, scikit-learn	Backend / server / ML code	MIT, BSD-3-Clause, BSD-3-Clause
ESCO v1.2.1, OpenAlex	Occupation/skill and research data	EC reuse 2011/833/EU, OpenAlex terms
RunPod SDK / LLM, Tavily, job adapters, Replit	AI/API/platform services	RunPod SDK MIT, model/provider terms require verification

Repository: github.com/weiiilunnnn/3am-sdc-hackathon (Apache-2.0 application code, third-party rights separate).

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, enter the affected member, agreement scope and any required employer approval.

-

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 25-10-2026